CUSTOMER CHURN PREDICTION FOR SYRIATEL

Overview

This project aims at analyzing customer churn for SyriaTel, a telecommunications company using machine learning, by building a classifier to predict whether a customer will (soon) stop doing business with the telecommunications company.

Business Understanding

- Churn is a major revenue risk since customer turnover impacts profitability. The main objectives of this project include :
- Predict customer most likely to churn
- Identify key features affecting customer churn
- Estimate revenue at risk
- Recommend retention strategies.

Data Understanding

- The data-set is from Kaggle and consist of 3,333 customers and 21 features.
- The target variable is Churn (Yes/No).

Data Preparation

- We imported the necessary libraries and ensured that the data-set is clean and reliable for further in-depth analysis. It entails checking for missing values, duplicates and outliers.

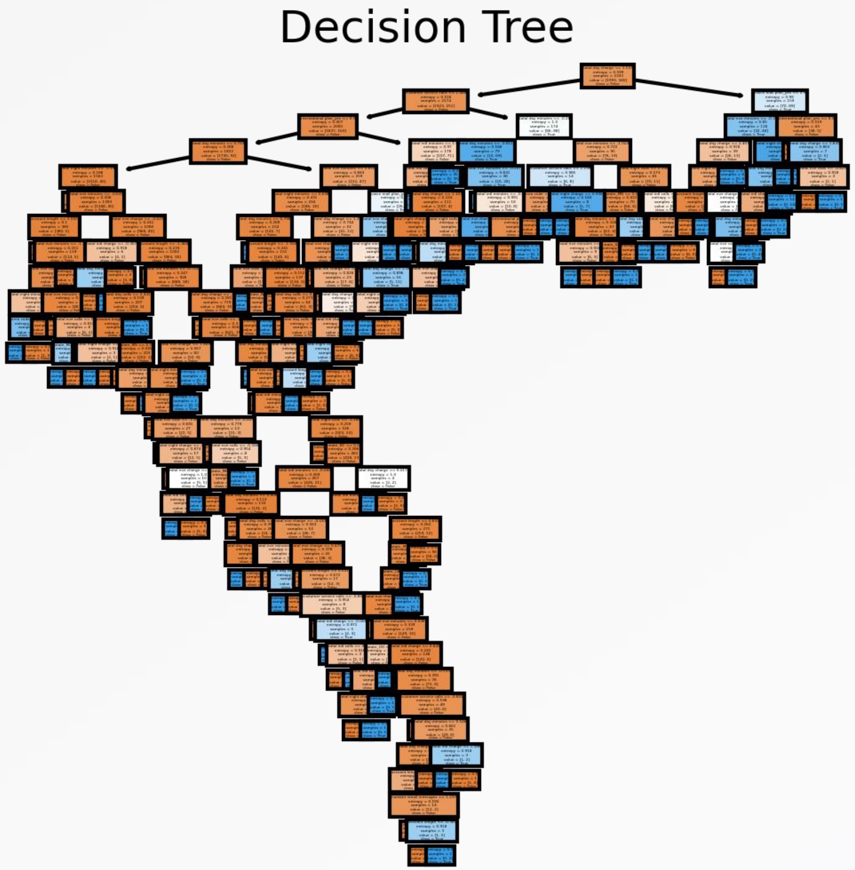
Modeling Approach

- The data-set was prepared for modelling by One Hot Encoding the Categorical features and scaling the numerical features.
- The data-set was then trained/ split where 70% of the data for training and 30% for testing.

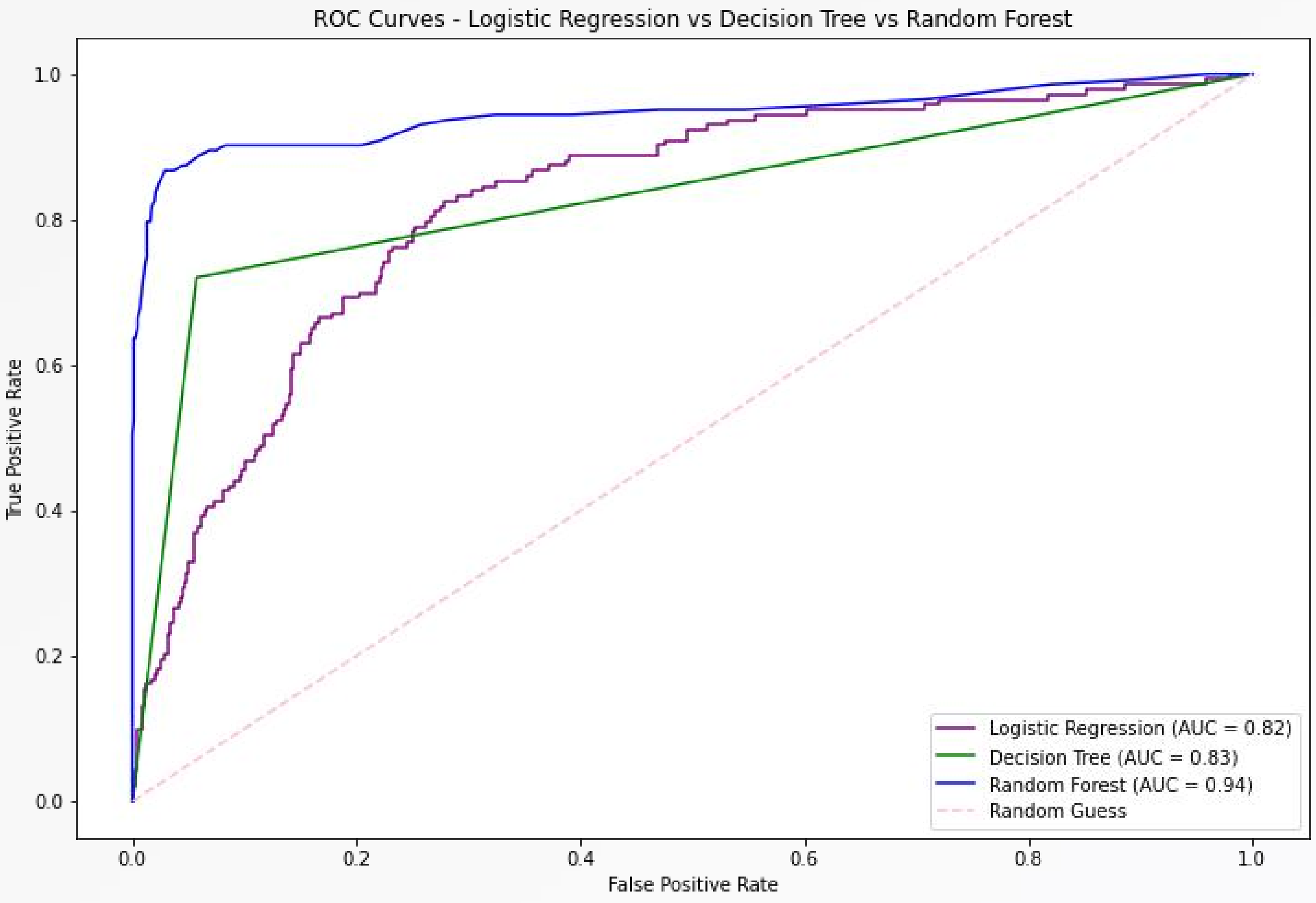
Model Testing

- We tested three models :
- Logistic Regression - Simple Baseline Model
- Decision Tree - Captures non- linear patterns
- Random Forest - A robust model And reduces over-fitting.

DecisionTree



ROC-AUC


ROC Curves - Logistic Regression vs Decision Tree vs Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.82)
Decision Tree (AUC = 0.83)
Random Forest (AUC = 0.94)
Random Guess

Machine Learning performance metrics

Use of Machine Learning metrics of success to evaluate performance of each model

- Accuracy,
- Precision,
- Recall,
- F1 score
- ROC_AUC

This will help us choose the best model to use in reducing churn by identifying at-risk customers early.

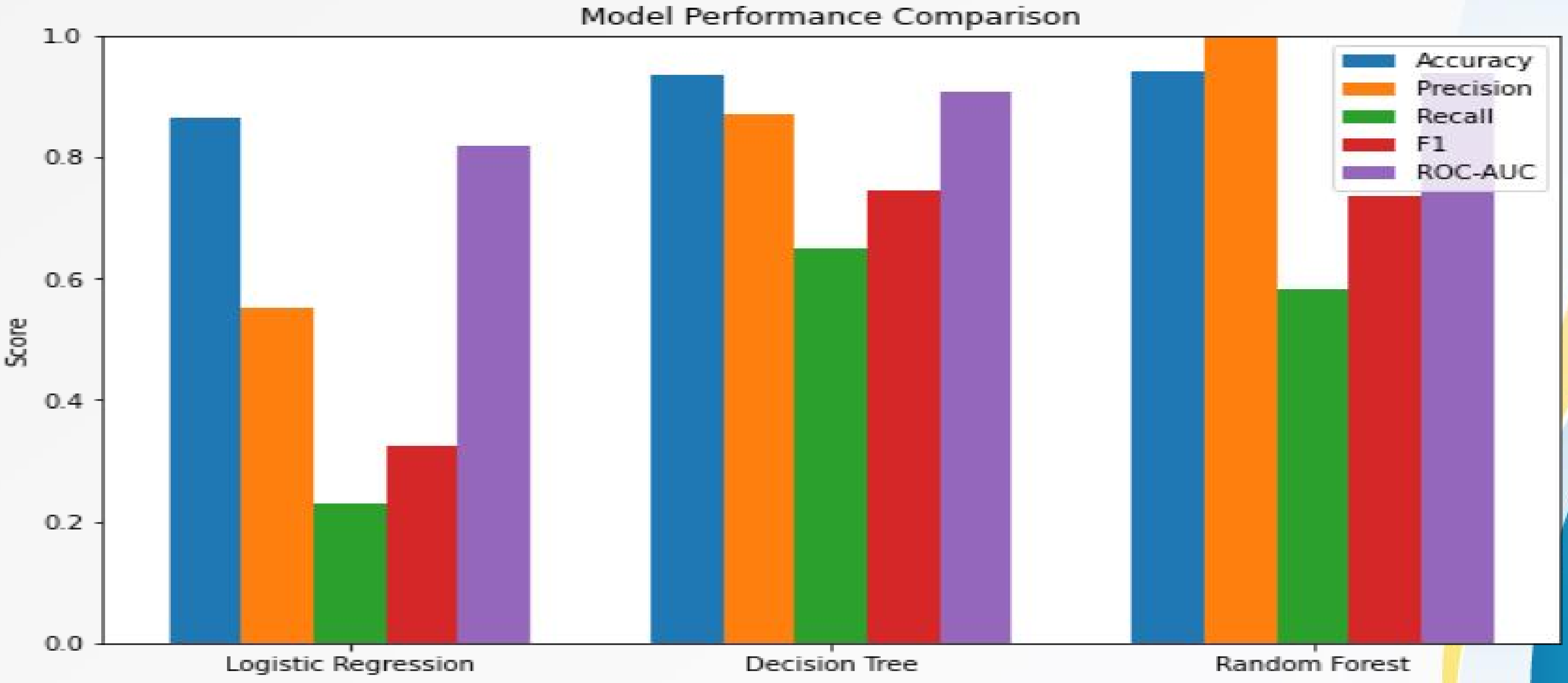
Evaluation of Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.863	0.55	0.230769	0.325123	0.818337
Decision Tree	0.936	0.869159	0.65035	0.744	0.907851
Random Forest	0.94	1	0.58042	0.734513	0.938617

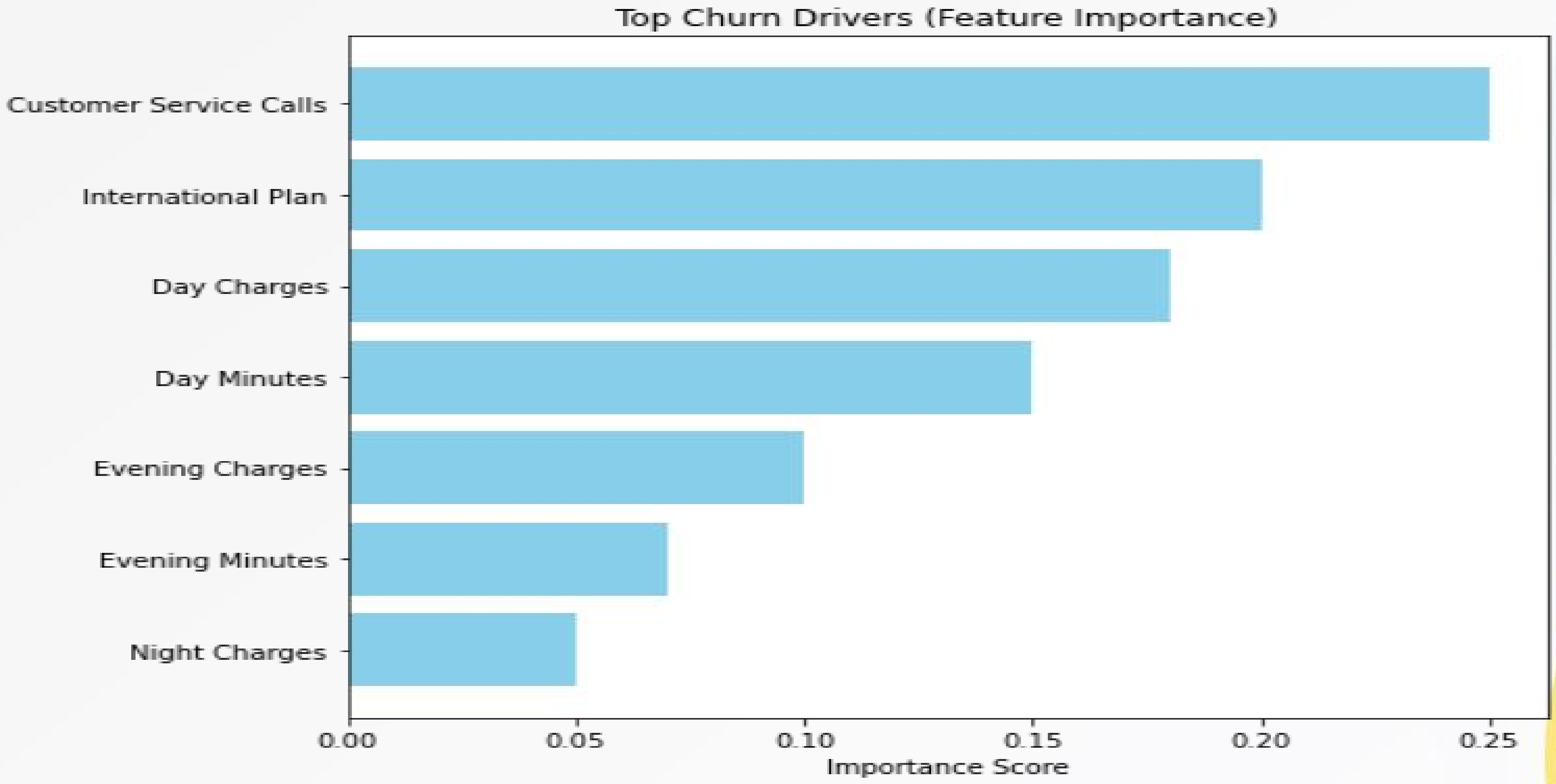
Summary

- Logistic Regression: struggles to detect churners,
- Decision Tree offers the best balance between precision and recall
- Random Forest is extremely precise but more conservative,catching fewer churners while being highly confident in its prediction.
- Thus, the best overall model is the Random Forest Classifier.

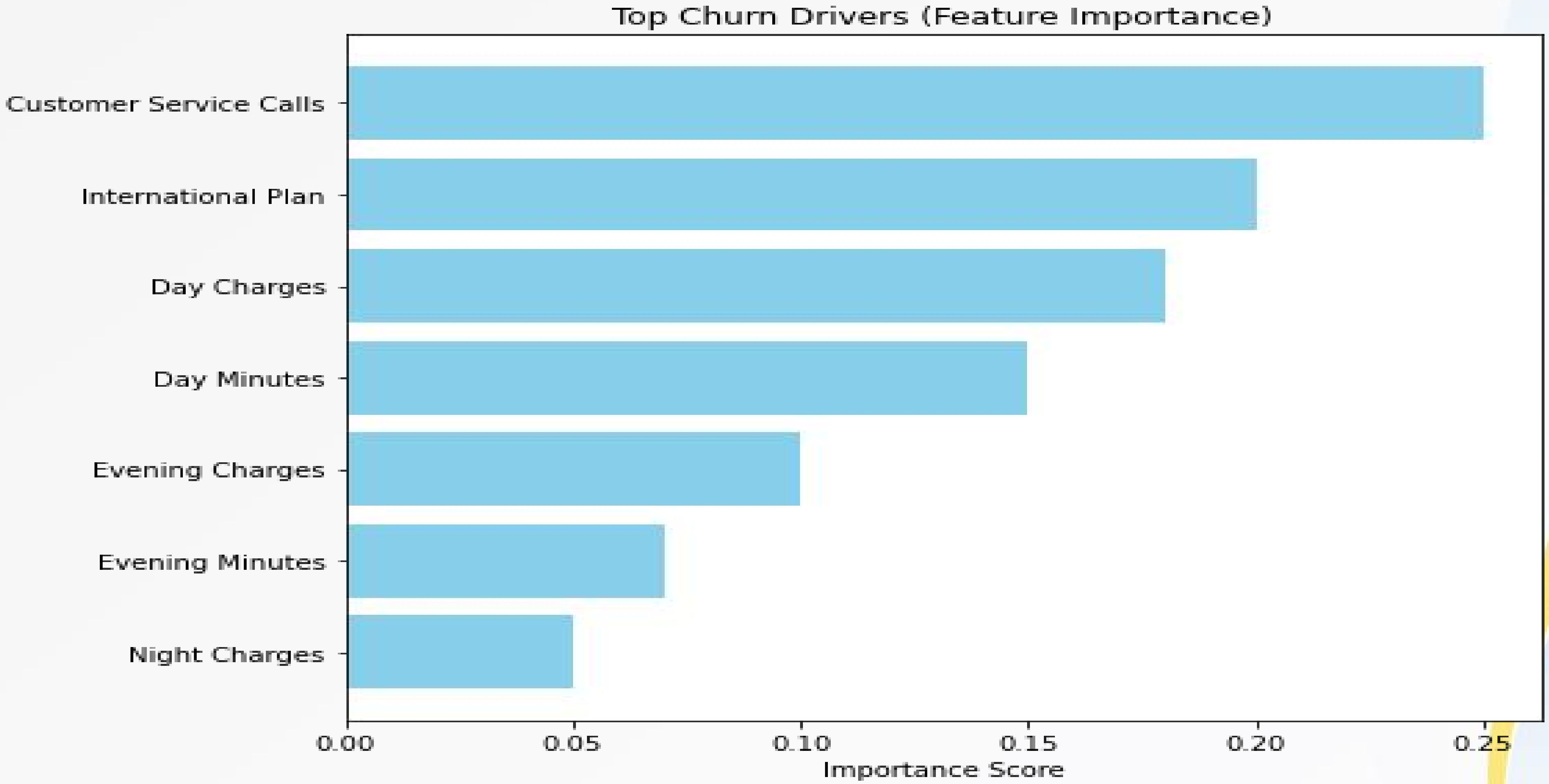
Model Performance Comparison



Random Forest in Identifying Top Churn Drivers.



Top Churn Drivers



Recommendations

- Feature Engineering: creation of new variables or combining the existing to improve prediction accuracy.
- Hyperparameter Tuning : Use of optimization techniques to fine tune model settings.
- Feature Importance: Random Forest shows strong performance and clear interpretability.
- Customer Retention Strategies: Direct interventions toward high-risk customers identified by all models.I.e service calls.

Next Steps:

- Expand the data-set to include demographics, tenure, service quality metrics
- Balance churn vs non-churn data for non biased predictions.
- Monitor model performance regularly
- Scale deployment across regions.

THANK YOU